Case E

Ms. Lee — Thoracic Outlet Syndrome | Working Dx: Neurogenic/Vascular Mixed TOS

Musculoskeletal Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Reduced first rib mobility
- Unable to perform overhead reaching at work without symptoms
- Difficulty sustaining repetitive upper limb tasks at end of shift
- Reduced right hand grip for packing/lifting
- Long-standing Type 1 Diabetes carries psychological burden — screen for diabetes distress and its interaction with pain experience

■■ RED FLAGS

- Persistent Hand Pallor / Cyanosis (Vascular Crisis)
- Suggests critical subclavian artery compression or thrombosis, especially dangerous in a diabetic patient with compromised vascularity. Risk of distal ischaemia and limb-threatening events.
- Action: Do NOT proceed with physical assessment. Urgent medical referral (ED or vascular surgery) if colour does not restore within minutes of position change.
- Progressive Muscle Wasting / Intrinsic Hand Weakness
- Indicates severe neurogenic TOS with lower brachial plexus involvement (C8/T1), potentially causing permanent neurological deficit if untreated.
- Action: Urgent referral to neurologist or vascular surgeon; imaging (MRI, Doppler USS, CT angiography) required before physiotherapy proceeds.
- Bilateral Upper Limb Symptoms / Cervical Myelopathy Signs
- Bilateral symptoms (gait disturbance, bowel/bladder changes) indicate spinal cord compromise rather than peripheral TOS, requiring urgent spinal investigation.

■ PATHOPHYSIOLOGY (KEY POINTS)

- Working Diagnosis: Thoracic Outlet Syndrome — Neurogenic/Vascular Mixed Type The thoracic outlet is a narrow triangular space bounded by the anterior and middle scalene muscles, the first rib, and the clavicle.
- The brachial plexus (C5–T1), subclavian artery, and subclavian vein all pass through this space, making any reduction in its dimensions clinically significant.
- This produces the mixed neurogenic-vascular presentation we see clinically.
- Tightness of pectoralis minor, confirmed on examination, creates a secondary compression point in the subcoracoid space.
- Lee elevates her arm overhead during warehouse tasks, this muscle shortens further, dragging the coracoid process down onto the neurovascular bundle and intensifying symptoms — explaining end-of-shift exacerbation.
- The intermittent arterial compromise produces vasospasm and transient ischaemia distal to the compression point, accounting for hand pallor and cyanosis.

■ DIFFERENTIAL DIAGNOSES

1. Thoracic Outlet Syndrome — Neurogenic/Vascular Mixed Type

2. Cervical Radiculopathy (C8–T1 Nerve Root Compression)

3. Diabetic Peripheral Neuropathy

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Key Observation Findings

Posture: Forward head posture, rounded/protracted right shoulder, possible elevated first rib on palpation, thoracic kyphosis Skin: Inspect right hand for pallor, cyanosis, or oedema; compare bilaterally Muscle bulk:

Test 1 — Roos Stress Test (EAST: Elevated Arm Stress Test)

What it measures Neurovascular compromise at the thoracic outlet under sustained provocative positioning; assesses both neurogenic and vascular TOS simultaneously. How to perform 1. Patient sits upright. Instruct them to abduct both shoulders to 90°, externally rotate maximally, and flex elbows to 90° (the "hands up" position). 2. Ask patient to open and close hands slowly and rhythmically for up to 3 minutes.

Test 2 — Adson's Maneuver (Scalene Compression Test)

What it measures Subclavian artery compression through the scalene triangle; assesses the arterial TOS component specifically. How to perform 1. Palpate and monitor the radial pulse on the affected right side. 2. Ask the patient to take a deep breath in and hold it (increases scalene tension).

■ TREATMENT PLAN

- I begin with consent because it establishes rapport, is ethically required, and signals a person-centred approach to the examiner.
- Test 3 — Upper Limb Neurodynamic Test 1 (ULNT1 / Median Nerve Bias)
- What it measures

• Mechanosensitivity of the median nerve and brachial plexus; detects neural tension and sensitisation along the neurodynamic chain from cervical spine to hand.
• 1. Patient supine. Depress the shoulder girdle (scapular depression).
• 2. Abduct the arm to 110° with elbow extended.
• 3. Supinate the forearm, extend the wrist and fingers fully.
• 4. Externally rotate the shoulder.
• 5. Add cervical lateral flexion away from the tested limb as a sensitisation step.
• 6. Compare bilaterally for symptom reproduction, quality, and available range.
• Positive finding
• Reproduction of paraesthesia/heaviness in right arm, reduced range compared to left, or symptom change with cervical sensitisation